INACTIVE FOR NEW DESIGN

MIL-V-25575B(USAF)

16 May 1983
Superseding
MIL-R-25575A(USAF)
20 March 1957

MILITARY SPECIFICATIONS

VALVE, REGULATING, FLUID PRESSURE, AIRCRAFT OXYGEN SYSTEM, GENERAL SPECIFICATIONS FOR

INACTIVE FOR NEW DESIGN after 16 May 1983 .

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for one type of pressure reducer as described herein.

2. APPLICABLE DOCUMENTS

2.1 Issures of documents. The following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-A-1034	-	Air, Compressed, for Breathing Purpose
BB-N-411	-	Nitrogen, Technical
PPP-T-60	-	Tape, Packaging, Waterproof
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)
PPP-B-636	-	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-P-7105	-	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

MIL-O-27210 - Oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27730 - Tape, Antiseize, Tetrafluorethylene, with Dispenser
MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited

STANDARDS

MILITARY

DOD-STD-100 - Engineering Drawing Practices
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-143 - Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-831 - Test Reports, Preparation of
MIL-STD-889 - Dissimilar Metals

2.1.2 Other Government documents, drawings and publications. The following other Government drawing forms a part of this specification to the extent specified herein.

Air Force Drawing

AF 8148650 - Cleaning, Inspection and Packaging of Breathing Oxygen Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.2).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray or atmospheric conditions likely to be met in storage or normal service.

3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

Source: https://assist.dla.mil -- Downloaded: 20260-20T06:43Z
Check the source to verify that this is the current version before use.

3.3.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

3.4 Design and construction. The oxygen pressure reducer shall be of the type suitable in all respects for use in aircraft oxygen systems. The reducer shall be designed to receive oxygen from the emergency supply at pressure from 50 to 2500 pounds per square inch gage, upon actuation of an emergency control, and to reduce the pressure to not less than 50 and no more than 80 pounds per square inch gage. The reducer shall be clean, dry and free from oil, grease, and other foreign matter. It shall contain no materials which will ignite when brought into direct contact with oxygen under high pressure.

3.4.1 Emergency control. The reducer shall be designed so that operation of a cord or cable will actuate a flow of oxygen from the emergency oxygen supply through the reducer. It shall be possible to reset this mechanism to again restrict the flow through the emergency inlet or facilitate reuse of the reducer. The length and size of cord or cable shall be as specified by the procuring activity.

3.4.2 Relief valve. The pressure reducer shall be provided with a safety device to insure complete protection to the distribution lines, fittings, and attached oxygen apparatus, and to prevent injury to the reducer itself resulting from malfunctioning due to any cause whatsoever. The safety device shall be so designed as to automatically restore normal operation of the reducer immediately after the condition causing the malfunction is eliminated.

3.5 Performance.

3.5.1 Relief valve. The relief valve shall prevent the outlet pressure from exceeding 130 pounds per square inch plus or minus 10 pounds per square inch and shall not release if the outlet pressure is below 85 pounds per square inch, when tested as specified in 4.6.2.

3.5.2 Pressure and flow. The pressure at the reducer outlet shall not show a variation greater than 8 pounds per square inch for flow rates up to 135 standard liters per minute, when tested as specified in 4.6.3.

3.5.3 Reduced pressure and flow. The pressure reducer shall meet the test specified in 4.6.5 when tested at inlet pressures of 1000 pounds per square inch, plus or minus 25 pounds per square inch; 500 pounds per square inch, plus or minus 25 pounds per square inch; 200 pounds per square inch, plus or minus 25 pounds per square inch; and 70 pounds per square inch, plus or minus 10 pounds per square inch.

3.5.4 Extreme temperature exposure. The reducer shall be capable of operating as specified in 4.6.5 within the temperature range of minus 65 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit, and plus 165 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit.

3.5.5 Leakage. The reducer shall withstand an internal pressure of 2000 pounds per square inch without leakage for not less than 2 minutes when tested as specified in 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 20260-20T06:43Z
Check the source to verify that this is the current version before use.

3.5.6 Vibration. The reducer shall be subjected to and meet the reducer pressure and flow test and the relief valve test after vibration as specified in 4.6.7.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part numbers.

3.7 Threads

3.7.1 Screw threads. Screw threads shall conform to MIL-S-7742.

3.7.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.7.3 Lubricant. When necessary to prevent seizing of the pipe threads, tape conforming to specification MIL-T-27730 shall be applied to male threads only.

3.8 Finish.

3.8.1 Aluminum materials. Aluminum and aluminum alloy parts shall be anodized in accordance with specification MIL-A-8625.

3.8.2 Ferrous materials. Ferrous materials shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or type II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coatings, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satifactory alternative processes can be used.

3.9 Maintenance. The design of the reducer shall be such as to accommodate to the greatest extent disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.10 Marking.

3.10.1 Identification of product. The reducer shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship. Workmanship shall be of the quality necessary to produce a product free from all defects which may adversely affect proper functioning in service.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection requirements specified herein shall be classified as follows:

a. Quality conformance inspection (see 4.3)

b. First article inspection (see 4.4)

4.3 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests

b. Sampling tests

4.3.1 Individual tests. Each reducer shall be subjected to the following tests, as described under 4.6, Test Methods, of this specification:

a. Examination of product

b. Relief valve

c. Pressure and flow

d. Leakage

4.3.2 Sampling plan and tests.

4.3.2.1 Sampling plan A. One reducer from each quantity of 10 for the first 100 reducers and one reducer from each 100 for each additional 1000, or fraction thereof, shall be subjected to the reduced pressure and flow test as described under 4.6, Test Methods.

4.3.2.2 Sampling plan B. One reducer from each quantity of 100 reducers and one reducer from each quantity of each additional 1000, or fraction thereof, shall be subjected to the tests listed below and described under 4.6 Test Methods:

a. Extreme temperature exposure

b. Vibration

4.3.2.3 Rejection and retest. When one or more items fail to meet the specified A and B tests, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

4.3.2.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found, and methods of correcting them.

4.4 First article testing.

4.4.1 First article test sample tested by the contractor. The contractor shall subject two reducers to the First Article tests specified in 4.4.4.

4.4.2 First article test report. After the contractor completes the First Article tests, he shall prepare a First Article test report according to MIL-STD-831 and furnish three complete copies of the report to the Government procuring activity.

4.4.3 First article test sample for the procuring activity. Along with the First Article test report, the contractor shall submit one sample to the Government procuring activity.

4.4.4 First article tests. First Article tests shall consist of all tests described under 4.6 Test Methods.

4.5 Test conditions

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25 ± 10°C (77 ± 18°F) and a relative humidity of not more than 80 percent. Where tests are made with atmospheric pressure or temperature different from the above values, proper allowance shall be made for the changes in instrument readings.

4.5.2 Gas. The high pressure gas used in testing the reducers shall be either oxygen conforming to MIL-O-27210, type I; nitrogen conforming to BB-N-411, class I, grade A; or air conforming to BB-A-1034, grade A or C. If nitrogen or air is used, appropriate density correction factors shall be applied to the flow meter used, not only to correct the effect on the meter itself, but also on the performance of the reducer with a lower density gas.

WARNING: Equipment used with oxygen must be kept clean and free of oil or other flammable compounds, and if even slightly contaminated, must be thoroughly cleaned with non-flammable solvent or cleaner before being used with oxygen. Explosion and fire may result unless these precautions are observed.

4.5.3 Orientation. The reducer shall operate satisfactorily as specified herein in any position in which it is mounted.

4.6 Test methods.

4.6.1 Examination of product. All completed reducers shall be carefully examined to determine conformance with this specification with respect to condition of finish, completeness of assembly, workmanship and any other visible defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

4.6.2 Relief valve. With a gage attached to the outlet port and a suitable pressure applied to the inlet port, the pressure at the outlet port shall be increased until the relief valve releases. The relief valve shall not release if the outlet pressure is below 85 pounds per square inch, but shall release at a pressure of not more than 130 pounds per square inch, plus or minus 10 pounds per square inch.

4.6.3 Pressure and flow. At an inlet pressure of 2000 pounds per square inch plus or minus 25 pounds per square inch, the outlet pressure shall be not less than 50 and not more than 80 pounds per square inch. Flow rates at the reducer outlet shall then be varied from zero to 135 standard liters per minute. The outlet pressure shall be recorded at various flow rates, and shall not show a variation in outlet pressure greater than 8 pounds per square inch for flow rates up to 135 standard liters per minute. The flows shall be 0, 10, 50, 100, and 135 liters per minute.

4.6.4 Leakage. The outlet end of the reducer shall be suitably plugged and pressure of 2000 pounds per square inch applied to the inlet end for not less than 2 minutes. Any evidence of leakage while at this pressure shall be cause for rejection.

4.6.5 Reduced pressure and flow. The pressure reducer shall meet the test specified in 4.6.3 except the 8 psig variation in outlet pressure, when tested at inlet pressures of 1000 pounds per square inch (psi) ± 25 psi; 500 psi ± 25 psi; and 200 psi ± 25 psi.

4.6.6 Extreme temperature exposure. The pressure reducer shall be subjected to a temperature of minus 65 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit, for a period of not less than 3 hours. While at this temperature, the reducer shall be tested in accordance with the reducer pressure and flow test and shall comply with all the requirements therein, except that the variation in outlet pressure shall not be greater than 10 pounds per square inch for flow rates up to 135 standard liters per minute. This test shall be repeated after the reducer has been subjected to a temperature of plus 165 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit, for a period of not less than 3 hours.

4.6.7 Vibration. The reducer shall be vibrated in a vertical plane at a double amplitude of not less than 0.018 and not more than 0.020 inch at frequencies between 500 and 2500 cycles per minute for a period of not less than 144 hours. The reducer shall then be subjected to and meet the reducer pressure and flow test and the relief valve test.

4.6.8 Preservation and packaging. Unless otherwise specified in the purchase order or contract, the supplier is responsible for the performance of all inspections required as listed in table III, MIL-P-116, for the method of preservation being accomplished.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A.

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

5.1.1.1 Cleaning. Reducer assemblies shall be cleaned in accordance with Air Force Drawing 8148650.

5.1.1.2 Drying. Reducer assemblies shall be dried in accordance with Air Force Drawing 8148650.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2) reducer assemblies shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the applicable requirements of that specification.

5.2 Packing. Packing shall be level B or C as specified (see 6.2).

5.2.1 Level B. The packaged reducer assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level C. The packaged reducer assemblies shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.3 Marking. Nomenclature tags, marked in accordance with MIL-STD-129 shall be securely attached to the unit package, showing title and part number without breaking the seals. In addition, each small unit package shall have a 2" X 6 1/2" or 3" X 10 1/2" label displayed on at least two sides of the container. The color of the labels shall be conspicuous green.

LIFE SUPPORT ITEM ALL OIL, GREASE, SHOP RESIDUE OR OTHER CONTAMINANTS HAVE BEEN REMOVED. DO NOT OPEN UNTIL READY FOR USE.

6. NOTES

6.1 Intended use. The pressure reducer is intended for use in aircraft where it is desired to reduce oxygen at pressures between 250 and 2500 pounds per square inch to a pressure range of 50 to 80 pounds per square inch at flow rates up to 135 standard liters per minute.

6.2 Ordering data. Procurement documents should specify the following:

a. First article requirements

b. Level of packaging and packing required

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description of the DOD Authorized Data List (TD-3). The data produced by this specification is DI-T-3718A, paragraph 1.b.

6.4 Identification of changes. The margins of this specification are not marked due to the extensiveness of the changes.

Custodian:
Air Force - 99

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 71

Project Number:
1660-F455

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

OC-ALC/MMEDO
Tinker AFB, OK 73145

Source: https://assist.dla.mil -- Downloaded: 202640-20T06:43Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:43Z
Check the source to verify that this is the current version before use.